A Controlled, End-to-End Comparison of EC2, ECS Fargate, and AWS Lambda

for Stateless Web Services

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github.com/Fighterforever/cs5296

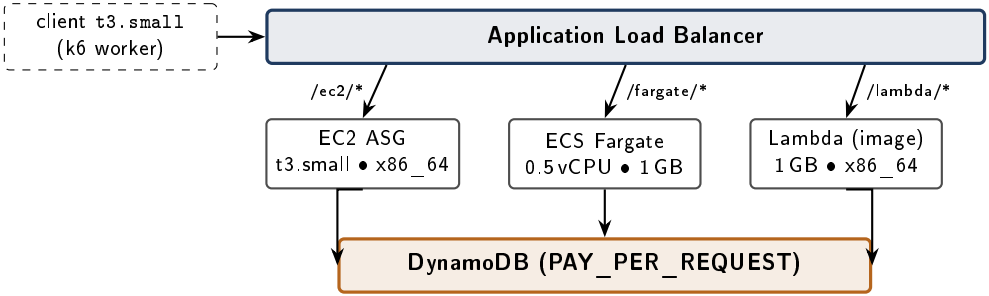
Three ways to run the same web service on AWS

EC2	ECS Fargate	AWS Lambda
Virtual machines.	Managed containers.	Functions on demand.
Predictable performance.	AWS handles the host.	No servers, no idle cost.
You handle scaling and OS upkeep.	*Your* task definition decides the size.	Cold-start penalty after idle.
Billed by the running hour.	Billed by the vCPU-second.	Billed per request + per GB-second.

Performance, elasticity, and cost rarely move in the same direction.

Most public studies compare different applications. We compare them apples to apples.

One container image, three runtimes, one ALB



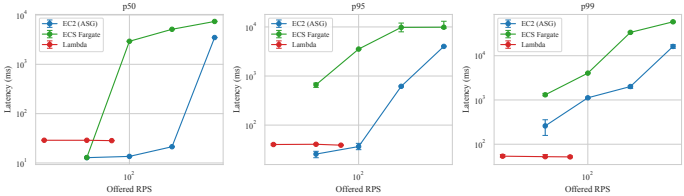
- Same OCI image (`shortlink:v2`) deployed unmodified to all three.
- Lambda runs the same image via the AWS Lambda Web Adapter, translating invocations into HTTP calls to localhost.
- Shared backend (DynamoDB) and shared frontend (ALB) confine measurable differences to the compute layer.

Four scenarios, driven by the same k6 harness

Steady-state	Constant arrival rate. EC2 and Fargate: 50/100/200/400 RPS. Lambda: 25/50/75 RPS. Recovers per-stage tail latency.
Step burst	One minute idle, three minutes at the target rate, one minute idle. Captures elasticity and time-to-steady-state.
Cold-start probe	Twelve sparse `/healthz` requests with 60-second gaps. Designed to characterise warm-vs-cold latency on Lambda.
Mixed R/W	70 % GET-redirect, 30 % POST-shorten at 200 RPS for three minutes. Approximates a real shortener mix.

All scenarios driven from a co-located `t3.small` k6 worker in `us-east-1a`; bootstrap 95 % CIs on every reported percentile.

Steady-state: Lambda is flat, EC2 has a fat tail, Fargate saturates

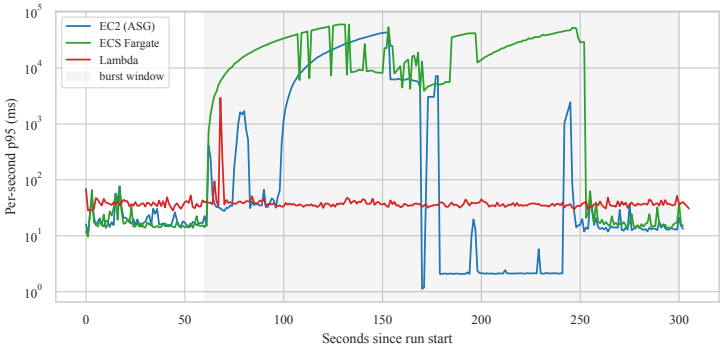


Lambda (1 GB)
p50 ≈ 29 ms, p95 ≈ 40 ms, p99 ≈ 52 ms across the entire band.

EC2 (t3.small ASG)
Tight 13 ms median up to 200 RPS. p95 goes from 26 ms to over 600 ms as the burstable-credit budget runs out.

Fargate (0.5 vCPU)
Saturates by 100 RPS. At 400 RPS, 6.9 % of requests hit the ALB idle timeout.

Elasticity: per-second p95 through a 3-minute burst

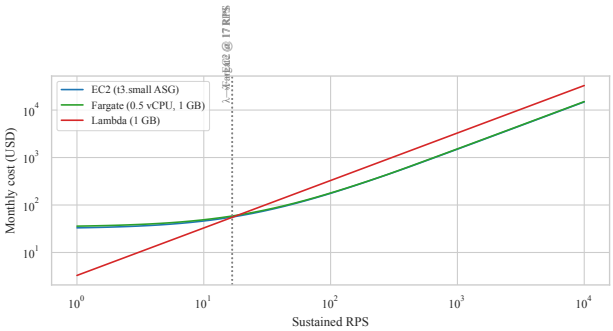


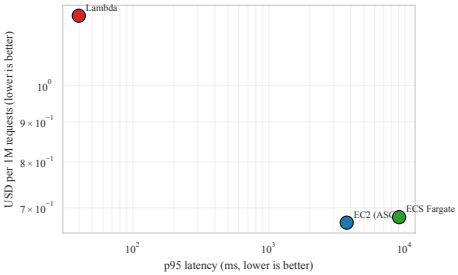
Lambda — tracks its steady envelope from the first second.

EC2 — takes about a minute for the second ASG instance to come online and rejoin the load balancer.

Fargate — the single task collapses; ECS service autoscaling at the default 30-second cooldown is too slow to recover.

Cost: a single break-even crossover near 20 RPS





- Below ∼20 RPS Lambda wins (no idle cost); above it pre-provisioned compute amortises enough.
- At 50 RPS Lambda is *Pareto-dominant* on both axes in our measured regime.

Live demo

Deployment, benchmark harness, and reproduction.

Reproducibility in five `make` targets

One repo, one command per phase

```
git clone https://github.com/Fighterforever/cs5296.git && cd cs5296
```

`make image`	build the OCI container image (`linux/amd64`, multi-runtime)
`make up`	`tofu apply`: 39 AWS resources in ∼7 minutes
`make bench`	drive the 4-scenario k6 matrix against the live ALB
`make analyze`	bootstrap stats, write 5 figures and 4 CSV summaries
`make report`	compile the LaTeX report end-to-end
`make nuke`	destroy + sweep stray ENI/EIP

- Infrastructure-as-code in OpenTofu/Terraform; portable to any AWS account.
- Analysis pipeline in Python; figures regenerate from raw k6 CSV in under a minute.
- Total measured AWS spend for one full matrix run: $< \$1$.

Three findings, two honest threats to validity

Findings

1. Lambda dominates up to ∼20 RPS — no-idle billing plus the most stable tail in the measured band.
2. EC2 keeps a tight median further than expected (200 RPS), but its tail blows out once the burstable-credit budget is depleted.
3. Fargate at the smallest size is *not* Lambda-with-containers — a single 0.5 vCPU task cannot absorb 100 RPS of Python.

Threats to validity

- Cold-start probe used 60-second gaps, too short to reliably reclaim Lambda containers (typical reclaim 5–15 min). All probe samples landed in the warm band; one genuine cold start (4.3 s) captured incidentally.
- New AWS account default Lambda concurrency was 10; this capped our Lambda peak at 75 RPS. A quota-increase ticket would lift it.

Thank you

Code, raw measurements, figures, and the report are public:

`github.com/Fighterforever/cs5296`